

CYBERSECURITY PORTFOLIO PROJECT

Secure Multi-Site Hospital Network

Network Design, Segmentation and Security Implementation

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Project Overview

This project presents the design and implementation of a secure multi-site hospital network using Cisco Packet Tracer. The network connects two separate hospital locations through an ISP/WAN infrastructure and uses a Site-to-Site IPsec VPN to secure communication between both sites.

The design applies multiple security controls, including VLAN segmentation, Access Control Lists (ACLs), Port Security, and unused port shutdown. The IPsec VPN uses AES encryption to protect data confidentiality, SHA to support data integrity, and a Pre-Shared Key for authentication between the two VPN peers.

The overall design follows a layered security approach to protect sensitive hospital resources, reduce unauthorized access, and secure data transmitted across the external network.

Project Scenario

The hospital operates across two separate locations. The first location contains a Manager PC and a Staff PC, while the second location contains two department PCs.

The Manager PC handles sensitive information and therefore requires additional protection. The network must provide reliable communication within each location and between both hospital sites while controlling access to sensitive resources and preventing unauthorized devices from connecting to the network.

Because communication between the two locations passes through an external ISP network, the traffic between both sites must also be protected against unauthorized access and data exposure.

Project Objective

The main objective of this project is to design and implement a secure and segmented network infrastructure for two hospital locations.

The project aims to:

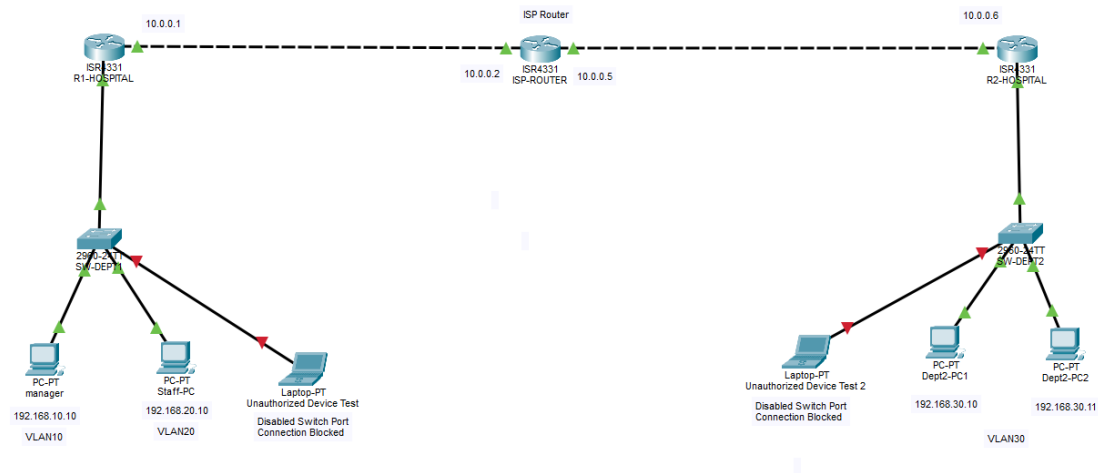
- Separate different users and departments using VLAN segmentation.
- Protect sensitive Manager PC resources using ACLs.
- Prevent unauthorized devices from accessing switch ports using Port Security.
- Disable unused switch ports to reduce unauthorized physical access.
- Secure communication between both hospital locations using a Site-to-Site IPsec VPN.
- Protect transmitted data using AES encryption.
- Support data integrity using SHA.
- Authenticate the VPN peers using a Pre-Shared Key.

Key Security Features

- VLAN Segmentation
- Access Control Lists (ACLs)
- Port Security
- Unused Port Shutdown
- Site-to-Site IPsec VPN
- AES Encryption
- SHA Integrity
- Pre-Shared Key Authentication

Network Design & Topology

Figure 1: Secure Multi-Site Hospital Network Topology



The network is divided into two separate hospital locations connected through an ISP/WAN infrastructure.

Site A contains the Manager PC and Staff PC, which are connected to SW-DEPT1 and routed through R1-HOSPITAL. The two devices are separated into different VLANs to provide logical network segmentation.

Site B contains two department PCs connected to SW-DEPT2 and routed through R2-HOSPITAL. Both devices are placed within VLAN 30.

The ISP Router represents the external WAN connection between the two hospital locations. R1-HOSPITAL and R2-HOSPITAL communicate through this WAN, while a Site-to-Site IPsec VPN is used to secure traffic transmitted between both sites.

The general communication path is:

Site A → R1-HOSPITAL → ISP Router → R2-HOSPITAL → Site B

IP Addressing & VLAN Design

The network uses separate VLANs and IP subnets to segment users and departments based on their roles and security requirements. VLAN 10 is assigned to the Manager network, VLAN 20 to the Staff network, and VLAN 30 to the second hospital department. /30 networks are used for the WAN links between the hospital routers and the ISP.

Segment	Device / Purpose	IP Address / Subnet	Gateway
VLAN 10	Manager-PC	192.168.10.10/24	192.168.10.1
VLAN 20	Staff-PC	192.168.20.10/24	192.168.20.1
VLAN 30	Dept2-PC1	192.168.30.10/24	192.168.30.1
VLAN 30	Dept2-PC2	192.168.30.11/24	192.168.30.1
WAN 1	R1 ↔ ISP	10.0.0.0/30	—
WAN 2	ISP ↔ R2	10.0.0.4/30	—

WAN Interfaces:

R1-HOSPITAL: 10.0.0.1 | ISP-R1: 10.0.0.2

ISP-R2: 10.0.0.5 | R2-HOSPITAL: 10.0.0.6

Security Controls Implementation

The network security design uses multiple controls to protect sensitive systems and reduce unauthorized access. Each security control addresses a different risk within the hospital network. The following sections describe how VLAN segmentation and Access Control Lists (ACLs) were implemented.

1. VLAN Segmentation

VLAN segmentation was used to logically separate users based on their roles and security requirements.

At Site A, the Manager PC was assigned to VLAN 10 using the **192.168.10.0/24** network, while the Staff PC was assigned to VLAN 20 using the **192.168.20.0/24** network.

The Manager PC was placed in a separate VLAN because it handles sensitive hospital information. This separation prevents the Manager and Staff devices from belonging to the same broadcast domain and allows security policies to be applied between the two networks.

At Site B, both department PCs were assigned to VLAN 30 using the **192.168.30.0/24** network.

The VLAN design provides logical network isolation while still allowing controlled communication between networks through the hospital routers.

VLAN	Network	Purpose
VLAN 10	192.168.10.0/24	Manager Network
VLAN 20	192.168.20.0/24	Staff Network
VLAN 30	192.168.30.0/24	Department 2 Network

2. Access Control List (ACL)

An Access Control List (ACL) was implemented to provide additional protection for the Manager PC located at 192.168.10.10.

The ACL restricts traffic from the Staff PC network to the Manager PC while allowing other authorized network traffic to continue normally.

This control is important because VLAN segmentation alone provides logical separation but does not automatically prevent communication between routed networks. The ACL adds an additional layer of access control by defining which traffic is allowed or denied.

In this project, traffic from the Staff PC at 192.168.20.10 to the Manager PC at 192.168.10.10 is blocked to reduce unnecessary access to sensitive hospital information.

Protected Host:

Manager-PC → 192.168.10.10

Blocked Traffic:

Staff-PC → Manager-PC

Allowed Traffic:

All Other Authorized Traffic

3. Port Security

Port Security was implemented on the access ports connected to end-user devices to prevent unauthorized devices from using switch ports assigned to legitimate hospital computers.

Each protected access port is configured to allow a maximum of one MAC address. If a different unauthorized device is connected to the protected port, a security violation occurs and the switch port is placed into a shutdown state.

At Site A, Port Security is applied to the ports connected to the Manager PC and Staff PC. At Site B, it is applied to the ports connected to Dept2-PC1 and Dept2-PC2. Uplink ports connected to the hospital routers are excluded from Port Security.

This provides an additional Layer 2 security control by restricting physical access to the internal hospital network.

Switch	Protected Ports	Connected Devices
SW-DEPT1	Fa0/1, Fa0/2	Manager-PC, Staff-PC
SW-DEPT2	Fa0/2, Fa0/3	Dept2-PC1, Dept2-PC2

4. Unused Port Shutdown

Unused switch ports were administratively disabled to reduce the risk of unauthorized physical access to the hospital network.

Only the ports required for legitimate devices and router uplinks remain active. All unused ports from Fa0/4 to Fa0/24 are placed in a shutdown state on both department switches.

This prevents an unauthorized user from connecting a device to an available switch port and immediately gaining physical access to the internal network.

Disabling unused ports is a simple but effective Layer 2 security measure that reduces the available attack surface of the network.

Switch	Active Ports	Disabled Ports
SW-DEPT1	Fa0/1 – Fa0/3	Fa0/4 – Fa0/24
SW-DEPT2	Fa0/1 – Fa0/3	Fa0/4 – Fa0/24

5. Site-to-Site IPsec VPN

A Site-to-Site IPsec VPN was implemented between R1-HOSPITAL and R2-HOSPITAL to protect communication between the two hospital locations.

The VPN creates a secure encrypted tunnel across the external ISP/WAN network. This allows traffic between the protected internal networks to travel securely even when it passes through an untrusted external network.

The VPN peers are:

R1-HOSPITAL: 10.0.0.1

R2-HOSPITAL: 10.0.0.6

The protected networks are:

Site A:

VLAN 10 → 192.168.10.0/24

VLAN 20 → 192.168.20.0/24

Site B:

VLAN 30 → 192.168.30.0/24

The VPN protects traffic between VLAN 10 and VLAN 20 at Site A and VLAN 30 at Site B.

6. VPN Encryption and Authentication

The IPsec VPN uses multiple security mechanisms to protect traffic transmitted between both hospital locations.

AES Encryption is used to provide confidentiality by encrypting data before it is transmitted across the external network. This helps prevent unauthorized parties from reading sensitive information.

SHA is used to support data integrity by helping verify that transmitted data has not been modified during transit.

A **Pre-Shared Key (PSK)** is used to authenticate the VPN peers. Both R1-HOSPITAL and R2-HOSPITAL must use the same shared secret before the secure VPN connection can be established.

Together, these mechanisms provide confidentiality, integrity, and authentication for communication between both sites.

Security Mechanism	Configuration	Purpose
VPN Type	Site-to-Site IPsec	Secure site communication
Encryption	AES	Data Confidentiality
Integrity	SHA	Data Integrity
Authentication	Pre-Shared Key	VPN Peer Authentication
VPN Peers	10.0.0.1 ↔ 10.0.0.6	R1-HOSPITAL ↔ R2-HOSPITAL

Security Testing Plan

A structured test plan was prepared to verify that the network infrastructure and implemented security controls operate as intended. Each test defines the security feature being evaluated, the testing procedure, the expected result, and the evidence required to confirm successful implementation.

Test ID	Test	Test Procedure	Expected Result	Evidence
T1	VLAN Verification	Check VLAN configuration on both switches	Devices are assigned to the correct VLANs	show vlan brief
T2	Allowed Connectivity	Send permitted traffic between authorized networks	Communication succeeds	Successful Ping
T3	ACL Protection	Ping Manager-PC from Staff-PC	Traffic is blocked by the ACL	Failed Ping + ACL Rule Verification
T4	Port Security	Connect an unauthorized device to a protected access port	Security violation occurs and the port shuts down	Port Security status
T5	Unused Port Shutdown	Connect a device to an administratively disabled port	No network connection is established	Interface status
T6	IPsec VPN	Generate traffic between Site A and Site B	Traffic passes through the encrypted VPN tunnel	IPsec SA + packet counters

Test 1 – VLAN Configuration Verification

The VLAN configuration was verified on both hospital switches using the show vlan brief command. On SW-DEPT1, VLAN 10 (MANAGER) is assigned to Fa0/1 and VLAN 20 (STAFF) is assigned to Fa0/2. On SW-DEPT2, VLAN 30 (DEPARTMENT2) is assigned to Fa0/2 and Fa0/3. The results confirm that the end devices are assigned to the correct VLANs and that logical network segmentation has been successfully implemented.

Figure2: VLAN 10 and VLAN 20 verification on SW-DEPT1

```
SW-DEPT1>show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2
10	MANAGER	active	Fa0/1
20	STAFF	active	Fa0/2
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
SW-DEPT1>
```

Figure 3: VLAN 30 verification on SW-DEPT2

```
SW-DEPT2>show vlan brief
```

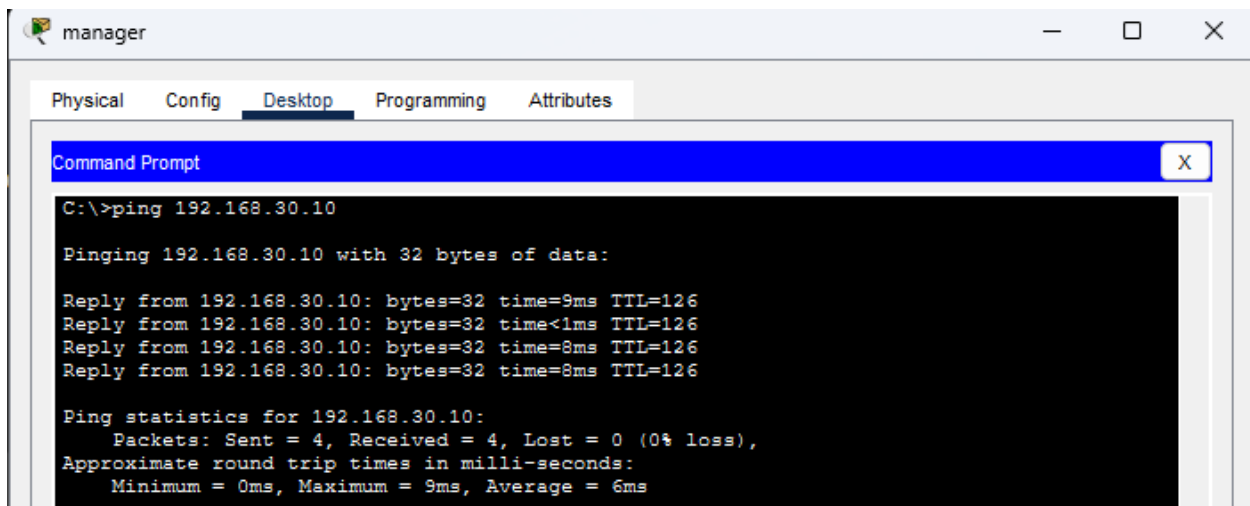
VLAN	Name	Status	Ports
1	default	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2
30	DEPARTMENT2	active	Fa0/2, Fa0/3
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
SW-DEPT2>
```

T2 – Allowed Connectivity

Allowed connectivity was tested by sending ICMP traffic from the Manager PC at Site A to Dept2-PC1 at Site B. The ping was successful, confirming that authorized traffic can pass between the two hospital locations and that the underlying routing and network connectivity are functioning correctly.

Figure 4: Successful Allowed Connectivity from Manager-PC to Dept2-PC1



The screenshot shows a network simulator window titled "manager". It has tabs for "Physical", "Config", "Desktop", "Programming", and "Attributes". The "Desktop" tab is active, displaying a "Command Prompt" window. The Command Prompt shows the execution of a ping command from C:\ to 192.168.30.10. The output indicates that the ping was successful, with 4 packets sent and received, 0% loss, and an average round trip time of 6ms.

```
C:\>ping 192.168.30.10

Pinging 192.168.30.10 with 32 bytes of data:

Reply from 192.168.30.10: bytes=32 time=9ms TTL=126
Reply from 192.168.30.10: bytes=32 time<1ms TTL=126
Reply from 192.168.30.10: bytes=32 time=8ms TTL=126
Reply from 192.168.30.10: bytes=32 time=8ms TTL=126

Ping statistics for 192.168.30.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 9ms, Average = 6ms
```

T3 – ACL Protection.

ACL protection was tested by sending ICMP traffic from the Staff PC at 192.168.20.10 to the protected Manager PC at 192.168.10.10. The ping request was blocked as expected. The ACL configuration on R1-HOSPITAL was then verified using the show access-lists command, confirming that traffic from the Staff PC to the Manager PC is denied while other authorized traffic is permitted.

Figure 5: ACL Blocking Staff-PC Access to Manager-PC

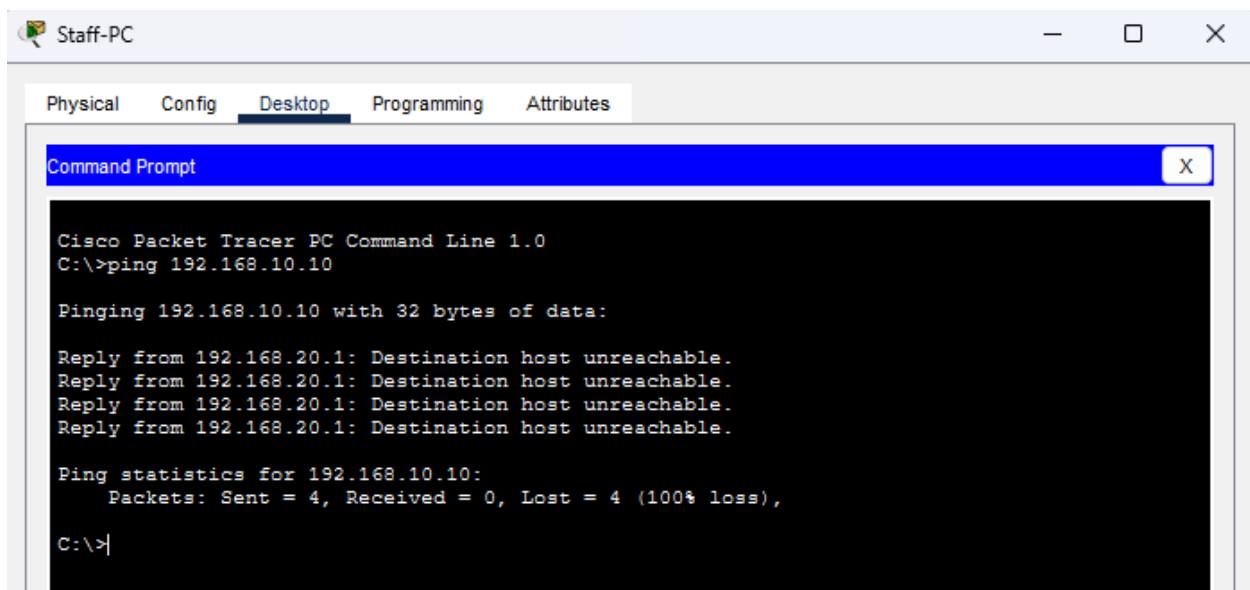


Figure 6: ACL Rule and Match Verification on R1-HOSPITAL

```
R1-HOSPITAL>
R1-HOSPITAL>
R1-HOSPITAL>enable
R1-HOSPITAL#show access-lists
Extended IP access list 110
 10 deny ip host 192.168.20.10 host 192.168.10.10
 20 permit ip any any
Extended IP access list 120
 10 permit ip 192.168.10.0 0.0.0.255 192.168.30.0 0.0.0.255
 20 permit ip 192.168.20.0 0.0.0.255 192.168.30.0 0.0.0.255
R1-HOSPITAL#
```

T4 – Port Security

Port Security was tested on the Fa0/1 interface of SW-DEPT1 by replacing the authorized Manager PC with an unauthorized laptop. The laptop was configured with a valid VLAN 10 IP address (192.168.10.50), but it was unable to access the network. The show port-security interface fa0/1 command confirmed that Port Security was enabled and that a security violation had occurred. The interface entered the Secure-shutdown state with a violation count of 1, successfully preventing the unauthorized device from accessing the hospital network.

Figure 7: Secure Sticky MAC Addresses Learned on SW-DEPT1

```
SW-DEPT1>en
SW-DEPT1>enable
SW-DEPT1#show port-security address
          Secure Mac Address Table
-----
Vlan      Mac Address      Type                Ports      Remaining Age
-----
  10      0002.4AA7.B8AC    SecureSticky        Fa0/1      -
  20      0002.17AC.E650    SecureSticky        Fa0/2      -
-----
Total Addresses in System (excluding one mac per port)    : 0
Max Addresses limit in System (excluding one mac per port) : 1024
SW-DEPT1#
```

Figure 8: Unauthorized Device Blocked by Port Security

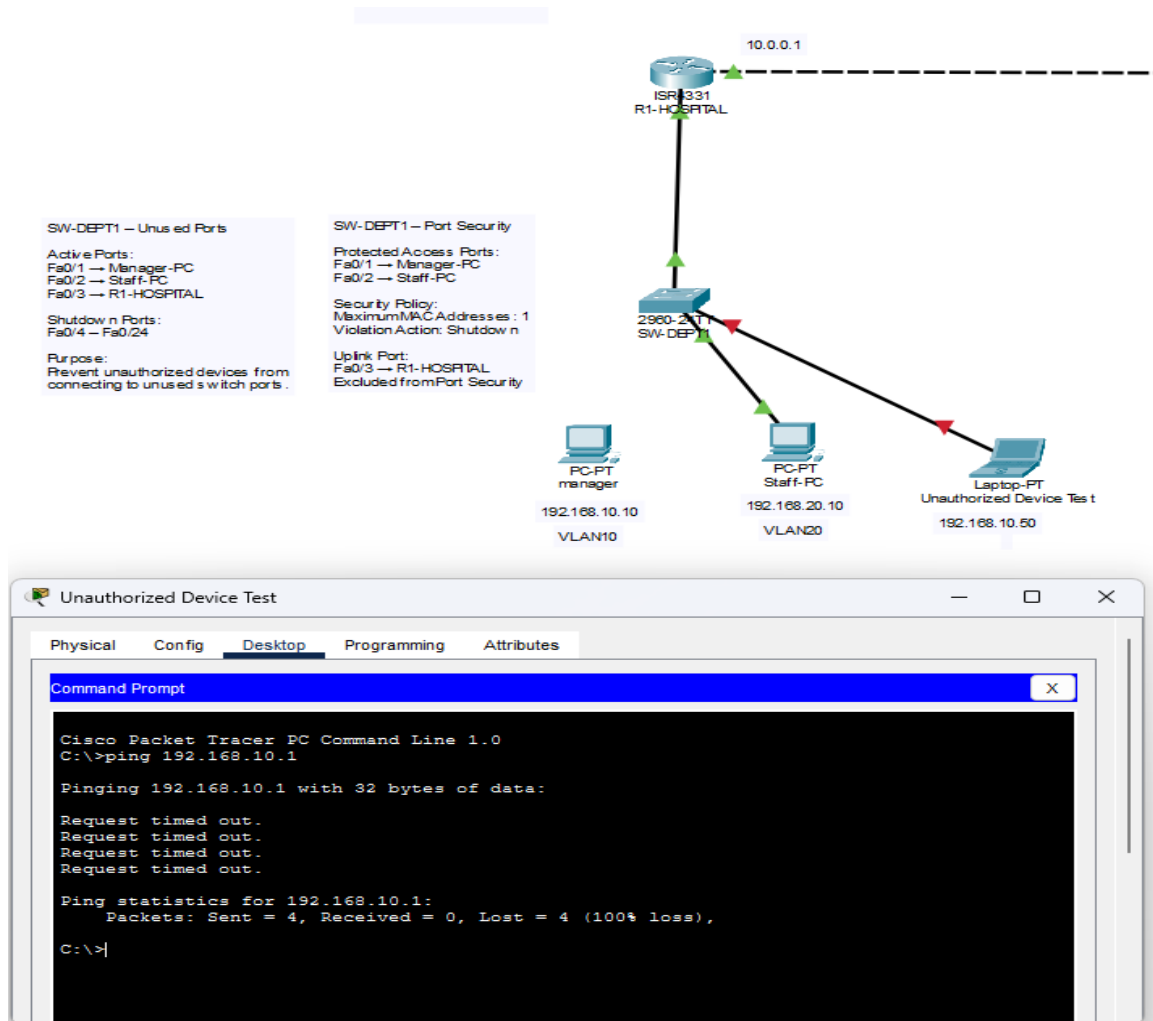


Figure 9: Port Security Violation and Secure-Shutdown Verification on SW-DEPT1

```
SW-DEPT1#
SW-DEPT1#
SW-DEPT1#
SW-DEPT1#show port-security interface fa0/1
Port Security           : Enabled
Port Status              : Secure-shutdown
Violation Mode           : Shutdown
Aging Time               : 0 mins
Aging Type               : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses    : 1
Total MAC Addresses      : 1
Configured MAC Addresses : 0
Sticky MAC Addresses     : 1
Last Source Address:Vlan : 00E0.A31D.214B:10
Security Violation Count : 1

SW-DEPT1#
```

T5 – Unused Port Shutdown

An unauthorized laptop was connected to the unused Fa0/4 interface on SW-DEPT2. The device was unable to establish network connectivity because the interface had been administratively disabled. The show interfaces fa0/4 command confirmed that the port was in an administratively down state. This verifies that unused switch ports are effectively disabled to prevent unauthorized physical access to the hospital network.

Figure 10: Unauthorized Device Connected to a Disabled Port on SW-DEPT2

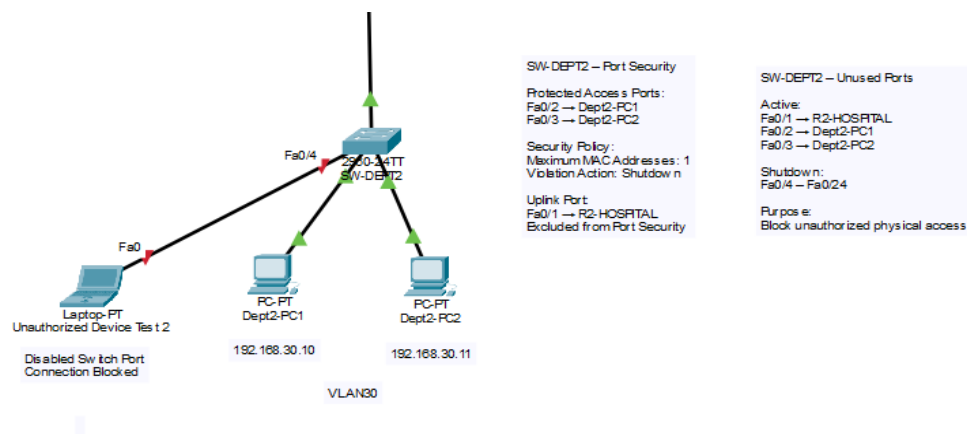


Figure 11: Administrative Shutdown Verification on SW-DEPT2 Fa0/4

```
SW-DEPT2>
SW-DEPT2>show interfaces fa0/4
FastEthernet0/4 is administratively down, line protocol is down (disabled)
Hardware is Lance, address is 0001.64a7.8304 (bia 0001.64a7.8304)
BW 100000 Kbit, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
Encapsulation ARPA, loopback not set
Keepalive set (10 sec)
Half-duplex, 100Mb/s
input flow-control is off, output flow-control is off
ARP type: ARPA, ARP Timeout 00:04:00
Last input 00:00:08, output 00:00:05, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
Queueing strategy: fifo
Output queue :0/40 (size/max)
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
  956 packets input, 193351 bytes, 0 no buffer
    Received 956 broadcasts, 0 runts, 0 giants, 0 throttles
  0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
  0 watchdog, 0 multicast, 0 pause input
  0 input packets with dribble condition detected
  2357 packets output, 263570 bytes, 0 underruns
--More--
```

T6 – Site-to-Site IPsec VPN Verification

Traffic was generated between Site A and Site B by sending ICMP packets from the Manager PC to Dept2-PC1. The connectivity test was successful with no packet loss. The show crypto isakmp sa command confirmed an active IKE security association between R1-HOSPITAL and R2-HOSPITAL with the state QM_IDLE. The show crypto ipsec sa command further confirmed that traffic between the protected networks was being encapsulated, encrypted, decapsulated, and decrypted through the IPsec tunnel. This verifies that the Site-to-Site IPsec VPN is operating successfully

Figure 12: Successful Connectivity from Manager-PC to Dept2-PC1

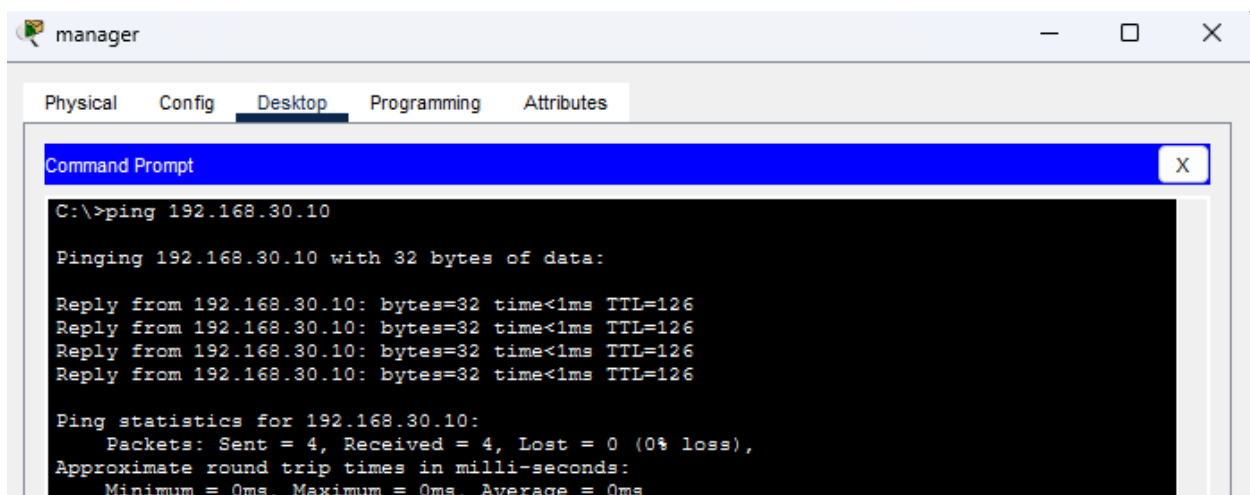


Figure 13: Active IKE Security Association Between R1-HOSPITAL and R2-HOSPITAL

```
R1-HOSPITAL>
R1-HOSPITAL>en
R1-HOSPITAL>enable
R1-HOSPITAL#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst          src          state          conn-id slot status
10.0.0.6     10.0.0.1     QM_IDLE        1072      0  ACTIVE

IPv6 Crypto ISAKMP SA

R1-HOSPITAL#|
```

Figure 14: IPsec Encrypted Traffic Verification on R1-HOSPITAL

```
R1-HOSPITAL#show crypto ipsec sa
interface: GigabitEthernet0/0/1
  Crypto map tag: HOSPITAL-MAP, local addr 10.0.0.1

  protected vrf: (none)
  local  ident (addr/mask/prot/port): (192.168.10.0/255.255.255.0/0/0)
  remote ident (addr/mask/prot/port): (192.168.30.0/255.255.255.0/0/0)
  current_peer 10.0.0.6 port 500
    PERMIT, flags={origin_is_acl,}
    #pkts encaps: 11, #pkts encrypt: 11, #pkts digest: 0
    #pkts decaps: 10, #pkts decrypt: 10, #pkts verify: 0
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
```

Test Results Summary

The following table summarizes the results of the security and connectivity tests performed on the hospital network. All implemented controls operated as expected during the verification process.

Test ID	Test	Actual Result	Status
T1	VLAN Verification	Devices were assigned to the correct VLANs	Pass
T2	Allowed Connectivity	Authorized communication between both sites was successful	Pass
T3	ACL Protection	Staff-PC access to Manager-PC was successfully blocked	Pass
T4	Port Security	Unauthorized device triggered a violation and Secure-shutdown	Pass
T5	Unused Port Shutdown	Unauthorized device could not connect through the disabled port	Pass
T6	IPsec VPN	Traffic between both sites was successfully encrypted through the VPN tunnel	Pass

The testing results confirm that the network design and implemented security controls are functioning as intended. VLAN segmentation successfully separates the internal networks, the ACL protects the Manager PC from unauthorized Staff access, Port Security prevents unauthorized devices from using protected switch ports, and unused ports are administratively disabled. In addition, the Site-to-Site IPsec VPN successfully provides encrypted communication between the two hospital locations.

Project Results

The project successfully achieved its main objective of designing and implementing a secure multi-site hospital network using Cisco Packet Tracer.

The network provides connectivity between two separate hospital locations while applying multiple layers of security. VLAN segmentation was successfully used to separate the Manager, Staff, and Department 2 networks. The ACL successfully restricted Staff-PC access to the protected Manager-PC, while Port Security prevented unauthorized devices from using protected switch ports.

Unused switch ports were administratively disabled to reduce unauthorized physical access. In addition, the Site-to-Site IPsec VPN successfully established secure communication between R1-HOSPITAL and R2-HOSPITAL through the ISP/WAN network.

Testing confirmed that traffic passing through the VPN tunnel was successfully encapsulated, encrypted, decapsulated, and decrypted. The implemented security controls operated as expected during verification.

Conclusion

This project demonstrates how multiple network security technologies can be combined to protect a small multi-site organization.

Rather than relying on a single security mechanism, the design uses layered security through VLAN segmentation, ACLs, Port Security, unused port shutdown, and Site-to-Site IPsec VPN protection.

The implementation provides logical network separation, access control, switch-level protection, and encrypted communication between remote locations. The testing process confirmed that the implemented controls function correctly and that the network can provide secure and reliable communication between both hospital sites.

Overall, the project demonstrates practical experience in network design, routing, switching, access control, Layer 2 security, and VPN implementation using Cisco Packet Tracer.

Future Improvements

The current design successfully meets the main security requirements of the project. However, several improvements could be implemented in a production environment to provide stronger security and monitoring capabilities.

- Deploy a dedicated Firewall between the internal network and external WAN.
- Implement IDS/IPS to detect and prevent suspicious network activity.
- Introduce centralized logging using Syslog or a SIEM platform.
- Implement stronger authentication methods instead of relying only on a Pre-Shared Key.
- Add network redundancy to reduce single points of failure.
- Implement centralized network monitoring and alerting.
- Apply additional ACL policies based on user roles and business requirements.
- Introduce secure wireless access using enterprise authentication if wireless connectivity is required.